

Chemical reactions

CHEMICAL REACTIONS ARE PROCESSES THAT CHANGE ONE SET OF SUBSTANCES INTO ANOTHER.

New bonds are made and existing ones are broken in a chemical reaction, which rearranges the atoms to form new substances.

SEE ALSO

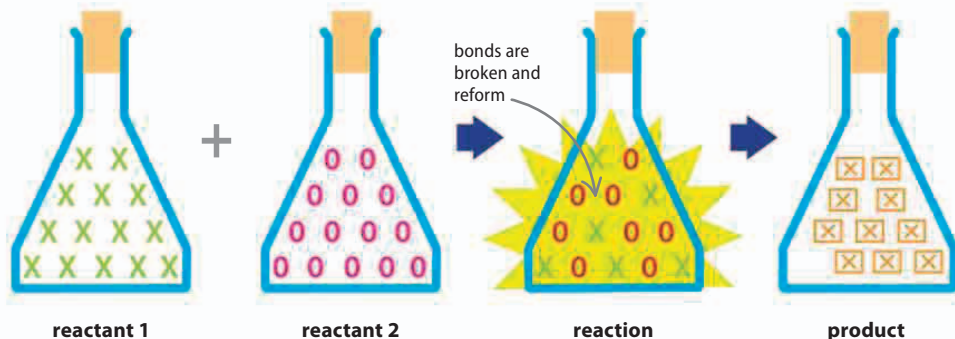
Combustion	130–131 >
Redox reactions	132–133 >
Energy and reactions	134–135 >
Rates of reaction	136–137 >
Catalysts	138–139 >
Reversible reactions	140–141 >

Start and end points

At the starting point of a chemical reaction are substances called reactants. Most reactions involve at least two reactants, although some involve only one reactant. The reactants can be compounds or pure elements. When they come into contact with each other, their ions and atoms are reorganized, resulting in the formation of a new set of substances, known as the products.

▷ Activating reaction

During the reaction, the bonds between atoms in the reactants rearrange, which results in the formation of the products. In this case two reactants have combined to form one product.



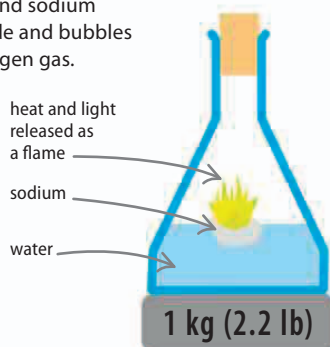
Self-rising flour
produces gas bubbles
by a chemical reaction
to make cakes light.

Conservation of matter

Atoms (or any other forms of matter) are neither created nor destroyed during chemical reactions. Every atom that was part of the reactants is present in the products, even if heat and flames are being released during the reaction. This principle is known as the conservation of matter.

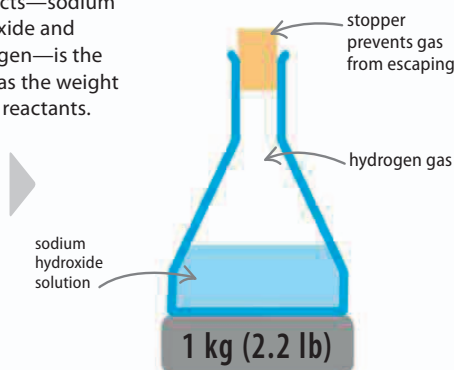
▷ Reactants

Sodium reacts with water to produce the compound sodium hydroxide and bubbles of hydrogen gas.



▷ Products

The weight of the products—sodium hydroxide and hydrogen—is the same as the weight of the reactants.



REAL WORLD

Sodas

The fizz of bubbles released when a sparkling drink is opened is produced by a decomposition reaction. Carbonic acid (H_2CO_3) dissolves in the water and breaks apart into carbon dioxide gas—which makes the refreshing bubbles—and more water.

